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EXPERIMENT 2

Aim: To build an end-to-end ETL pipeline: extract data from an API, transform it and load it into a database.

Theory:

Introduction to ETL

ETL stands for Extract, Transform, Load, which is a process used in data integration and data management. It's typically used to prepare data for analysis or reporting. Here's a breakdown of each step:

1. Extract:
 - This is the first step, where data is pulled from various sources such as databases, APIs, flat files, or cloud services. The goal is to gather raw data from diverse and possibly disparate systems.
2. Transform:
 - Once data is extracted, it undergoes transformation, which involves cleaning, structuring, and converting it into a desired format. This step might involve filtering data, performing calculations, normalizing values, handling missing data, or combining information from multiple sources.
3. Load:
 - The final step is to load the transformed data into a target system, which could be a data warehouse, database, or data lake. The data is now ready for analysis, reporting, or further processing.

ETL is critical for preparing data for business intelligence tools, ensuring that the data is accurate, consistent, and usable for decision-making.

Implementation of ETL pipeline

To implement an ETL pipeline, you typically need to write scripts or use tools designed for data extraction, transformation, and loading. Below is a simple outline of how to implement an ETL pipeline, followed by an example using Python and libraries like pandas for transformation, SQLAlchemy for database interaction, and requests for API data extraction.

Steps to Implement an ETL Pipeline

1. Extract Data:
 - Connect to the data source (API, database, or file).
 - Extract raw data from the source.
2. Transform Data:



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- Clean the data by handling missing values, duplicates, or irrelevant information.
 - Convert data into a structured format for easier processing (e.g., flattening nested data).
 - Perform any calculations or data aggregation as needed.
3. Load Data:
- Load the transformed data into a target system (e.g., a database or data warehouse).

Screenshots:

1) Connection Establishment:

```
In [1]: pip install mysql-connector-python

Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: mysql-connector-python in c:\users\djsce.student\appdata\roaming\python\python311\site-packages (9.2.0)
Note: you may need to restart the kernel to use updated packages.

In [1]: import mysql.connector

mydb = mysql.connector.connect(
    host = 'localhost',
    user = 'root',
    password = 'pass@123',
    database = 'dbase'
)

print(mydb)

<mysql.connector.connection.MySQLConnection object at 0x00000182C9FC6910>
```

2) Retrieving data from database

```
In [2]: mycursor = mydb.cursor()

In [3]: mycursor.execute('SELECT * FROM employee')

In [4]: myresult = mycursor.fetchall()

for x in myresult:
    print(x)

(1, 'Ashok', 'Badlapur', 'BCOM', 101, 'Sales', 'Manager', datetime.date(2000, 9, 22), datetime.date(2023, 8, 11), None)
(2, 'Shravan', 'Nagpur', 'BSC', 102, 'system', 'HOD', datetime.date(1982, 8, 18), datetime.date(2005, 2, 7), None)
(3, 'Adhur', 'Andheri', 'BFM', 103, 'system', 'Senior_Manager', datetime.date(1977, 8, 2), datetime.date(2004, 1, 2), None)
```

3) Extracting data from API



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EXTRACTING DATA FROM API

```
[10]: pip install requests

Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: requests in c:\programdata\anaconda3\lib\site-packages (2.31.0)
Requirement already satisfied: charset-normalizer<4,>=2 in c:\programdata\anaconda3\lib\site-packages (from requests) (2.0.4)
Requirement already satisfied: idna<4,>=2.5 in c:\programdata\anaconda3\lib\site-packages (from requests) (3.4)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\programdata\anaconda3\lib\site-packages (from requests) (2.0.7)
Requirement already satisfied: certifi<=2017.4.17 in c:\programdata\anaconda3\lib\site-packages (from requests) (2024.2.2)
Note: you may need to restart the kernel to use updated packages.

[11]: import requests
import mysql.connector
from datetime import datetime

# 1. Fetch data from the NewsAPI
api_url = 'https://newsapi.org/v2/everything?q=tesla&from=2025-01-04&sortBy=publishedAt&apiKey=b83c45154a7546238ef532c75720999f'
response = requests.get(api_url)

# Check if the request was successful (status code 200)
if response.status_code == 200:
    data = response.json() # Convert the response to a dictionary
else:
    print(f"Error fetching data: {response.status_code}")
    exit()

# 2. Connect to MySQL Database
db = mysql.connector.connect(
    host="localhost", # Database server address
    user="root", # Database username
    password="pass@123", # Database password
    database="dbase" # Database name
)

cursor = db.cursor()

# 3. Create the table (if it doesn't exist already)
cursor.execute("""
CREATE TABLE IF NOT EXISTS XYZ_NEWS (
    id INT AUTO_INCREMENT PRIMARY KEY,
    title VARCHAR(255),
    description TEXT,
    url VARCHAR(1024),
    published_at DATETIME
)
""")

# 4. Insert data into the MySQL database
articles = data.get('articles', []) # Extract articles list from the response
for article in articles:
    title = article.get('title')
    description = article.get('description')
    url = article.get('url')
    published_at = article.get('publishedAt')

    # Convert the 'publishedAt' value to a MySQL-compatible DATETIME format
    if published_at:
        # Remove the 'Z' at the end and replace 'T' with a space
        published_at = published_at.replace('T', ' ').replace('Z', '')
        # Optionally, you can also convert it to a datetime object
        try:
            published_at = datetime.strptime(published_at, '%Y-%m-%d %H:%M:%S')
        except ValueError:
            print(f"Invalid date format: {published_at}")
            continue

    # Insert data into the table
    cursor.execute("""
INSERT INTO XYZ_NEWS (title, description, url, published_at)
VALUES (%s, %s, %s, %s)
""", (title, description, url, published_at))

# Commit the transaction
db.commit()

# 5. Close the database connection
# cursor.close()
# db.close()

print("Data inserted successfully!")

Data inserted successfully!
```

4) Displaying extracted data



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DISPLAYING FROM EXTRACTED DATA

```
[14]: mycursor = db.cursor()
mycursor.execute("select * from XYZ_NEWS")
myresult = mycursor.fetchall()

for x in myresult:
    print(x)
```

```
(36, 'USAID turmoil: DOGE reps take over offices, senior officials placed on leave - NewsBreak', 'Amid ongoing turmoil inside the U.S. Agency for International Development, sources say that Department of Government Efficiency staffers have moved to take', 'https://www.newsbreak.com/news/3788205413842-usaid-turmoil-doge-reps-take-over-offices-senior-officials-placed-on-leave?noAds=1&f=app_share&s=a2', datetime.datetime(2025, 2, 3, 0, 56, 53))
(37, 'USAID turmoil: DOGE reps take over offices, senior officials placed on leave - NewsBreak', 'Amid ongoing turmoil inside the U.S. Agency for International Development, sources say that Department of Government Efficiency staffers have moved to take', 'https://www.newsbreak.com/share/3788205413842-usaid-turmoil-doge-reps-take-over-offices-senior-officials-placed-on-leave?_f=app_share&pd=04pDrK4n&lang=en_US&send_time=1738543982&trans_data=%7B%22platform%22%3A%22c%22%3A%22224.39.1.3%22%22languages%22%3A%22en%22%7D&sep=ns_push_exp_rt_bucketv12-v2%2Cns_foryou_blend_exp_25q1-v2%2Cns_foryou_model_exp_25q1-v8%2Cns_foryou_rank_exp_25q1-v4%2Cns_foryou_recall_exp_25q1-v9&s=i2', datetime.datetime(2025, 2, 3, 0, 53, 34))
(38, 'USAID turmoil: DOGE reps take over offices, senior officials placed on leave - NewsBreak', 'Amid ongoing turmoil inside the U.S. Agency for International Development, sources say that Department of Government Efficiency staffers have moved to take', 'https://www.newsbreak.com/share/3788205413842-usaid-turmoil-doge-reps-take-over-offices-senior-officials-placed-on-leave?_f=app_share&pd=0JN8T2pw&lang=en_US&send_time=1738543591&trans_data=%7B%22platform%22%3A%22c%22%3A%22225.4.0.27%22%22languages%22%3A%22en%22%7D&sep=ns_foryou_blend_exp_25q1-v1%2Cns_foryou_rank_exp_25q1-v9%2Cns_foryou_recall_exp_25q1-v9%2Cns_foryou_model_exp_25q1-v6%2Cns_push_exp_rt_bucketv12-v8', datetime.datetime(2025, 2, 3, 0, 47, 3))
(39, 'Singapore says Nvidia's astounding local sales don't mean it's the source of DeepSeek's GPUs', 'Singapore says Nvidia's astounding local sales don't mean it's the source of DeepSeek's GPUs\nPLUS: Chinese bus lanes put Tesla in a tangle; India drops electronics tariffs; Samsung worries about
```

Conclusion:

Thus we successfully built an end-to-end ETL pipeline: extracted data from an API, transformed it and loaded it into a database.